

Human Granulocytic Anaplasmosis (HGA): Comprehensive Disease Characterization Report

Target Disease: Human Granulocytic Anaplasmosis **MONDO ID:** MONDO:0005118 **Category:** Infectious Disease (tick-borne zoonotic bacterial infection) **Causative Agent:** *Anaplasma phagocytophilum* (human-active "Ap-ha" variant), NCBI Taxonomy ID: 948

Summary

Human granulocytic anaplasmosis (HGA) is an acute, usually self-limiting tick-borne febrile illness caused by the obligate intracellular gram-negative bacterium *Anaplasma phagocytophilum*. It is transmitted principally by *Ixodes scapularis* (eastern/midwestern United States) and *Ixodes pacificus* (western United States), with small mammals — chiefly the white-footed mouse *Peromyscus leucopus* — serving as enzootic reservoirs. Only a single "human active" (Ap-ha) genetic variant of the bacterium causes human disease. HGA is fundamentally an **infectious** disease with **no human genetic etiology**: there are no causal genes, inheritance patterns, pathogenic germline variants, or vaccines. The dominant human "risk factors" are environmental and behavioral (tick exposure in endemic regions), with advanced age and immunosuppression predicting severe disease.

The pathogen has a unique biology: it is one of the few bacteria that survives and replicates inside **neutrophils**, the body's primary antibacterial effector cells. It accomplishes this through an elaborate program of subversion — entering via the PSGL-1/sialyl-Lewis^x receptor complex, then deploying Type IV secretion system (T4SS) effectors **AnkA** (which epigenetically silences the NADPH-oxidase gene *CYBB* via HDAC1 recruitment, blocking the respiratory burst) and **Ats-1** (which enters host mitochondria to block apoptosis and hijacks autophagy for nutrient acquisition). Crucially, the tissue injury and clinical severity of HGA are driven not by direct bacterial cytotoxicity — bacterial burdens are low relative to disease severity — but by the **host immune response**, specifically an IFN- γ /STAT1-driven immunopathology. In its most extreme form this dysregulation manifests as secondary hemophagocytic lymphohistiocytosis (HLH).

Clinically, HGA presents as a nonspecific febrile illness (fever in 88.5% of cases) accompanied by thrombocytopenia (71.8%), abnormal liver injury tests (66.7%), and leukopenia (49.8%). It is diagnosed by whole-blood PCR and IFA serology, with blood-smear morulae showing low sensitivity in early disease. **Doxycycline is rapidly curative first-line therapy for all ages**, with fever typically subsiding within one day; rifampin is the alternative in pregnancy. Prognosis is generally good (lethality ~3.0%, sequelae ~2.1%), though complications occur in ~40% of cases and advanced age plus immunosuppression predict severe/hospitalized disease. Prevention rests entirely on tick-bite avoidance, as no vaccine exists. The same bacterium naturally infects horses, dogs, cats, sheep, cattle, and goats, making HGA part of a broader multi-host zoonosis with important veterinary parallels.

Key Findings

Finding 1: HGA is a tick-borne infection of neutrophils presenting as a febrile illness with cytopenias

A systematic review of HGA cases established the core clinical and laboratory phenotype. HGA "primarily presents as an unspecific febrile illness (88.5% of the cases) often accompanied by thrombocytopenia (71.8% of the cases), abnormal liver injury tests (66.7% of the cases), and leukopenia (49.8% of the cases)" ([PMID: 39102427](#)). Complications occurred in 40.5% of cases (acute renal failure 9.8%, multi-organ failure 7.5%, ARDS 6.3%), while "sequelae are rare (2.1% of the cases) and lethality is low (3.0% of the cases)."

This phenotypic signature — fever plus the triad of thrombocytopenia, transaminitis, and leukopenia — is the diagnostic fingerprint of HGA and reflects the pathogen's tropism for hematopoietic and granulocytic lineages. The nonspecific nature of the presentation is a central clinical challenge, since it overlaps with many other acute febrile illnesses and other tick-borne diseases (Lyme disease, babesiosis, ehrlichiosis).

Suggested HPO terms: Fever (HP:0001945), Thrombocytopenia (HP:0001873), Elevated hepatic transaminase (HP:0002910), Leukopenia (HP:0001882), Myalgia (HP:0003326), Headache (HP:0002315), Acute kidney injury (HP:0001919).

Finding 2: *A. phagocytophilum* survives in neutrophils by subverting bactericidal functions and delaying apoptosis

The paradox of HGA is that the pathogen thrives inside the very cell type designed to destroy it. The bacterium "infects and actively grows in neutrophils by employing an array of mechanisms to subvert their bactericidal activity. These include its ability to inhibit phagosome-lysosome fusion, to suppress respiratory burst and to delay the apoptotic death of neutrophils" (PMID: 17275372).

An Affymetrix microarray study of infected human polymorphonuclear leukocytes (PMNs) demonstrated the apoptosis-evasion mechanism directly: "ingestion of *A. phagocytophilum* failed to trigger the neutrophil apoptosis differentiation program that typically follows phagocytosis and ROS production" (PMID: 15879137). Normally, neutrophils undergo a programmed apoptotic death shortly after phagocytosis; by suppressing this program, the bacterium extends the lifespan of its intracellular niche, buying time for replication.

Suggested GO terms: negative regulation of apoptotic process (GO:0043066), negative regulation of respiratory burst (GO:0060264), phagosome-lysosome fusion (GO:0090385).

Suggested CL term: neutrophil (CL:0000775).

Finding 3: Type IV secretion effectors AnkA and Ats-1 drive intracellular subversion

The molecular machinery of subversion is the T4SS and its two best-characterized effectors. **AnkA** (Ankyrin A) "enters the granulocyte nucleus, binds stretches of AT-rich DNA and alters transcription of antimicrobial defence genes, including down-regulation of *CYBB*" (PMID: 25996657). AnkA recruits histone deacetylase 1 (HDAC1) to the *CYBB* (gp91phox/NADPH oxidase) promoter, causing histone H3 deacetylation and epigenetic silencing — directly disabling the enzyme responsible for the neutrophil respiratory burst. This is a striking example of a bacterial effector functioning as a host transcriptional/epigenetic regulator.

Ats-1 (Anaplasma translocated substrate 1) operates at the mitochondria: it "inhibited etoposide-induced cytochrome c release from mitochondria, PARP cleavage, and apoptosis in mammalian cells" (PMID: 20174550). Ats-1 additionally hijacks autophagy: it "binds Beclin 1, a subunit of the class III PI3K and Atg14L, and it nucleates autophagosomes" (PMID: 23197835), redirecting autophagosomal cargo to feed the bacterium. Multi-omics work further shows Ats-1 up-regulates respiratory-chain subunits (NDUFB5, NDUFB3, NDUFS7, COX6C, SLC25A5), enhancing host ATP production and inhibiting apoptosis to support bacterial replication.

Suggested GO terms: modulation by symbiont of host process (GO:0044003), negative regulation of host apoptotic process, autophagosome assembly (GO:0000045), histone deacetylation (GO:0016575).

Finding 4: Doxycycline is highly effective; advanced age and immunosuppression predict severe disease

"Treatment with doxycycline shows a rapid response, with the fever subsiding in the majority of patients within one day of starting treatment" (PMID: 39102427). This rapid defervescence is so characteristic that a prompt response to empiric doxycycline supports the diagnosis.

Prognostic stratification comes from a 465-case Mayo Clinic cohort (2011–2021), in which 33% of patients were hospitalized. "Hospitalized patients (n = 153, 33%) were more likely to be older (median age of 71 vs 61; $P \leq .001$) and immunocompromised (17% vs 7%; $P \leq .001$)" (PMID: 40176262). Additional risk factors for hospitalization included altered mental status, higher absolute neutrophil count, and comorbidities. Notably, coinfection (e.g., with *Borrelia burgdorferi* or *Babesia microti*) did not impact mortality or hospitalization in this cohort.

Suggested NCIT term: Doxycycline (NCIT:C731).

Finding 5: Entry via PSGL-1/sialyl-Lewis^x and persistence via msp2(p44) antigenic variation

The molecular entry mechanism is well defined: "P-selectin glycoprotein ligand-1 (PSGL-1) and the tetrasaccharide sialyl Lewis x (sLe(x)), which caps the PSGL-1 N-terminus, are confirmed *A. phagocytophilum* receptors" (PMID: 18485118). PSGL-1 N-terminus-mediated entry is Syk-dependent and promotes optimal delivery of the Anka effector, linking receptor engagement to the downstream subversion program.

Persistence and immune evasion are achieved through antigenic variation of the immunodominant major surface protein: "*A. phagocytophilum* utilizes gene conversion to shuffle approximately 100 functional pseudogenes into a single expression cassette of the msp2(p44) gene, which encodes the major surface antigen, major surface protein 2 (Msp2)" (PMID: 22859615). In chronically infected reservoir woodrats, 60 unique expression-site variants emerged over the course of infection — a continuously moving antigenic target that frustrates the antibody response.

Finding 6: HGA is an emerging *Ixodes*-borne zoonosis; only the Ap-ha variant is human-pathogenic

HGA is transmitted primarily by *Ixodes scapularis* (eastern/midwestern US) and *I. pacificus* (western US), with reservoirs in small mammals, chiefly *Peromyscus leucopus* (white-footed mouse) and eastern chipmunks. Critically, only one variant infects humans: studies reference "the zoonotic variant Ap-ha (human active) of the bacterium *Anaplasma phagocytophilum*" (PMID: 41016325). Approximately 90–100% of infected reservoir small mammals in southeastern Canada carried Ap-ha.

The disease burden is substantial and rising. US hospitalization data show that "Lyme disease was the most common cause, accounting for 65% of hospitalizations (171,328 admissions), followed by ehrlichiosis/anaplasmosis (46,446)" over 2002–2021 (PMID: 41003548), with tick-borne disease hospitalizations increasing 2.5-fold. The burden concentrates in the Northeast (52.9% of TBD hospitalizations), peaks in July, and affects males slightly more (53.9%). Rare non-tick transmission occurs: "Although usually transmitted via tick bite, HGA may rarely also be acquired through transfusion" (PMID: 25385549) — and leukoreduction does not reliably prevent transfusion transmission — as well as perinatally.

Finding 7: HGA tissue injury is immunopathologic, driven by IFN- γ /STAT1 signaling

A defining and mechanistically important feature of HGA is that disease severity is disproportionate to bacterial burden, pointing to an immune-mediated pathology. "IFN- γ , is necessary for innate immunity and plays an important role in the induction of severe histopathology in *A. phagocytophilum*-infected mice, horses and humans" (PMID: 23278812). The downstream signaling node is STAT1: an "increase in phosphorylated Stat1 (pStat1) correlated significantly with IFN- γ production and inflammatory tissue injury," with phosphorylated STAT1 markedly increased by day 7 post-infection in infected mice.

This immunopathologic model reframes HGA: the pathogen sets off a host inflammatory cascade (via TLR2/NF- κ B and IFN- γ /STAT1) that produces the observed organ injury. It is reinforced by the observation, in CNS involvement, that "CSF abnormalities did not correlate with neurologic severity, suggesting a cytokine-mediated process rather than direct central nervous system infection" (PMID: 42230277).

Suggested GO terms: interferon-gamma-mediated signaling pathway (GO:0060333), inflammatory response (GO:0006954), response to interferon-gamma (GO:0034341).

Finding 8: Diagnosis relies on PCR and serology; blood-smear morulae have low early sensitivity

"The laboratory diagnosis is most frequently serological—evidence of antibody by indirect immunofluorescence assay (IFA) and detection of DNA by polymerase chain reaction (PCR), or microscopy evidence—Giemsa stain of blood smears (morulae in granulocytes or monocytes)" (PMID: 20077398). Morulae are intracytoplasmic microcolonies of the bacterium — the pathognomonic but insensitive microscopic finding.

A Korean hospital cohort quantified this insensitivity: "Of the 18 patients who underwent peripheral blood (PB) smear test, only one (5.6%) had morulae" (PMID: 34035378). Because both morulae detection and IFA (which requires seroconversion) are insensitive in early disease, whole-blood PCR is the primary early diagnostic tool, and empiric doxycycline should not await confirmation.

Finding 9: Doxycycline is first-line for all ages; rifampin is the alternative in pregnancy

In a case series of six pregnant women with HGA, disease was non-fulminant and "all treated patients had excellent responses to rifampin or doxycycline therapy. Perinatal transmission was documented in 1 neonate, who responded well to treatment" (PMID: 17682993), with no long-term sequelae in offspring at a mean follow-up of 21 months. A transfusion-associated case in a pregnant patient was successfully treated with rifampin (PMID: 25385549). Doxycycline remains the drug of choice across all systematic reviews and all age groups; rifampin is reserved for pregnancy and doxycycline intolerance.

Suggested NCIT terms: Doxycycline (NCIT:C731), Rifampin (NCIT:C692).

Finding 10: *A. phagocytophilum* is a multi-host zoonotic pathogen (granulocytic anaplasmosis / tick-borne fever)

The same bacterium causes disease across many mammalian species. Genetic typing (ankA, groEL, MLST) shows "the *A. phagocytophilum* strains found infecting cats are the same as those that cause disease in humans, dogs and horses" (PMID: 37803346). In naturally PCR-positive horses, hematological abnormalities occurred in 95%, dominated by thrombocytopenia (86%) and anemia (52%) — paralleling the human thrombocytopenia phenotype (PMID: 36436292). In ruminants the disease is called tick-borne fever (TBF): "By itself TBF does not cause high

mortality rates but infected animals are more susceptible to other secondary infections, pregnant animals may abort and there is a severe reduction in milk yield in dairy cattle" ([PMID: 17275372](#)).

Suggested NCBI Taxonomy identifiers for affected species: *Homo sapiens* (9606), *Equus caballus* (9796), *Canis lupus familiaris* (9615), *Felis catus* (9685), *Ovis aries* (9940), *Bos taurus* (9913), *Capra hircus* (9925), *Peromyscus leucopus* (10041).

Finding 11: Severe anaplasmosis can trigger secondary HLH, a treatable immune-dysregulation complication

The immunopathologic model reaches its extreme in secondary HLH. Case reports "present two cases of severe anaplasmosis that progressed to secondary hemophagocytic lymphohistiocytosis (HLH). This severe immune dysregulation syndrome has an extremely high mortality, but anaplasmosis represents one of the few treatable underlying etiologies" ([PMID: 32723647](#)). One "anaplasmosis-induced HLH successfully treated with a combination of doxycycline, steroids, and anakinra (an IL-1 receptor antagonist)" — demonstrating that this primarily immune-mediated complication responds to both antimicrobial therapy and immunosuppression, and reinforcing the cytokine-driven mechanism.

Suggested NCIT terms: Anakinra (NCIT:C1594), Corticosteroid therapy. **Suggested HPO term:** Hemophagocytosis (HP:0012156).

Finding 12: Integrated model — pathogen-driven neutrophil subversion produces host IFN- γ /STAT1 immunopathology, reversible with doxycycline

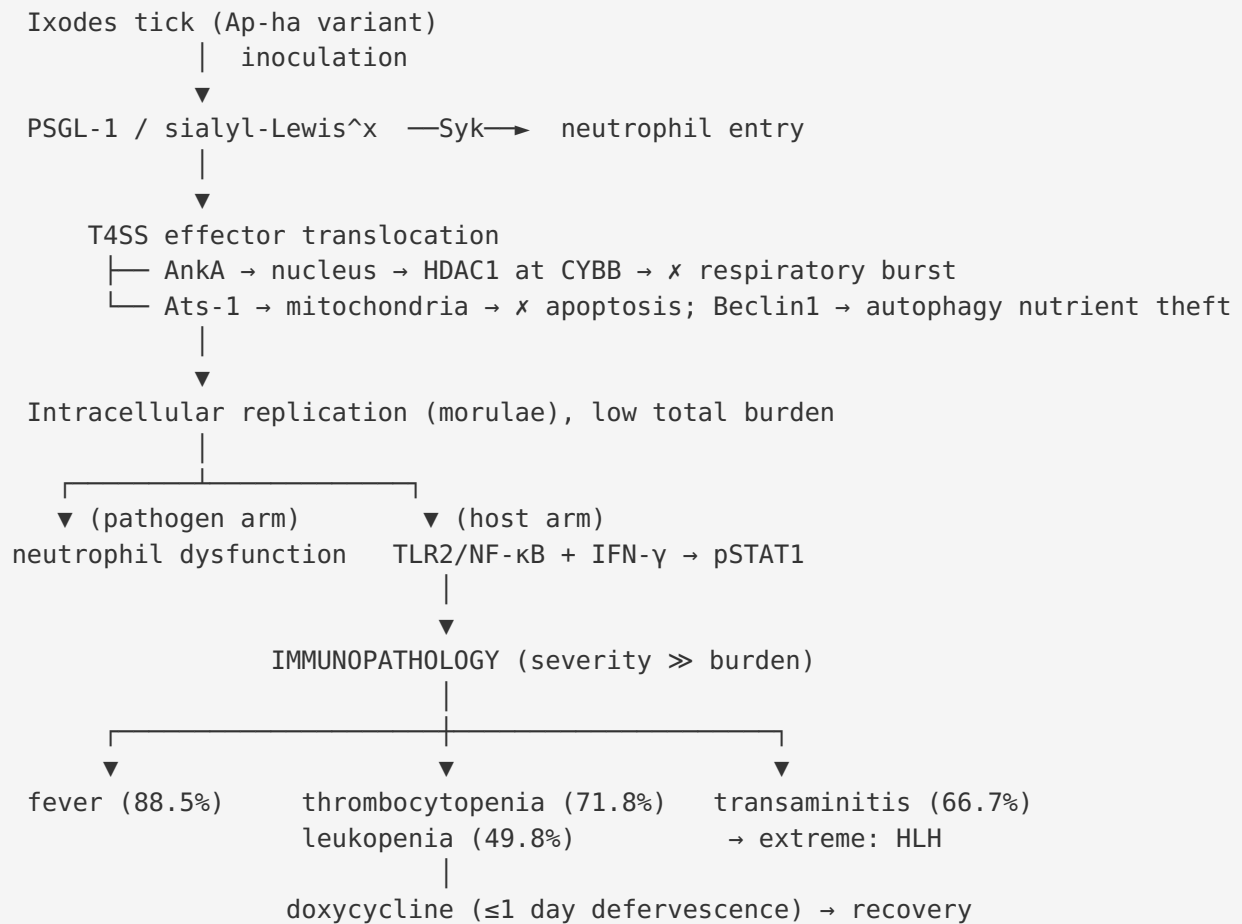
Synthesizing all findings, HGA follows a coherent causal chain from tick inoculation of the Ap-ha variant to a febrile cytopenic illness that is fully reversible with doxycycline. The upstream arm is pathogen-driven (receptor entry $\rightarrow$ T4SS effectors $\rightarrow$ neutrophil subversion); the downstream arm is host-driven (IFN- γ /STAT1 immunopathology $\rightarrow$ clinical disease, and in extreme cases HLH). Clinically this yields fever (88.5%), thrombocytopenia (71.8%), transaminitis (66.7%), and leukopenia (49.8%) with low lethality (3.0%), and doxycycline produces defervescence within $\sim$ 1 day. There is no human causal gene, inheritance pattern, or vaccine; prevention is tick-bite avoidance. The two anchoring facts of this synthesis are the clinical output ([PMID: 39102427](#)) and the immunopathology driver ([PMID: 23278812](#)).

Mechanistic Model / Interpretation

Ordered causal chain (initiating exposure → clinical manifestation)

1. An *Ixodes scapularis* or *I. pacificus* tick carrying the **Ap-ha (human-active)** variant of *A. phagocytophilum* bites a human and inoculates the bacterium. (*demonstrated — vector/variant epidemiology*)
2. Circulating bacteria bind **PSGL-1** capped by **sialyl-Lewis^x** on neutrophils and myeloid precursors, **leading to** Syk-dependent internalization into a host-derived vacuole. (*demonstrated in vitro*)
3. Internalization **results in** assembly of the Type IV secretion system and translocation of effectors into the host cell. (*demonstrated*)
4. **AnkA** traffics to the nucleus, binds AT-rich DNA at the *CYBB* promoter, and recruits **HDAC1**, **causing** histone H3 deacetylation and epigenetic silencing of *CYBB* (gp91phox) — this **blocks the NADPH-oxidase respiratory burst**. (*demonstrated*)
5. In parallel, **Ats-1** enters host **mitochondria** and inhibits cytochrome c release and PARP cleavage, **blocking apoptosis**; it also binds **Beclin 1/Atg14L** to nucleate autophagosomes and divert nutrients to the bacterium. (*demonstrated in vitro/cell models*)
6. Suppressed oxidative killing + delayed apoptosis + nutrient acquisition **result in** unchecked intracellular bacterial replication and formation of morulae. (*demonstrated*)
7. **Branch — immune sensing**: bacterial ligands engage **TLR2 → NF-κB** and drive **IFN-γ** production; IFN-γ signaling **leads to** STAT1 phosphorylation (pSTAT1). (*demonstrated in mouse/comparative models; inferred in humans*)
8. Elevated **IFN-γ/pSTAT1 causes** inflammatory, immunopathologic tissue injury that is **disproportionate to bacterial burden** — producing fever, hepatic transaminitis, and organ dysfunction. (*demonstrated in animal models; strongly inferred in humans*)
9. Consumption/sequestration and marrow effects **result in thrombocytopenia and leukopenia**; the febrile cytopenic syndrome is the clinical readout. (*demonstrated clinically; mechanism partly inferred*)
10. **Extreme branch**: in a subset (older/immunocompromised), unchecked cytokine activation **leads to** secondary **HLH**. (*demonstrated in case reports*)
11. **Doxycycline** halts bacterial protein synthesis, **collapsing** the effector-driven subversion and interrupting the cytokine cascade, **resulting in** defervescence within ~1 day and full recovery in the great majority. (*demonstrated clinically*)

Schematic



Upstream vs downstream

Layer	Mechanism	Direction	Evidence type
Vector/variant	<i>Ixodes</i> transmission of Ap-ha	Most upstream	Epidemiology (human)
Entry	PSGL-1/sLe ^x /Syk	Upstream	In vitro
Subversion	AnkA (CYBB silencing), Ats-1 (anti-apoptosis, autophagy)	Upstream	In vitro / cell models
Immune sensing	TLR2/NF-κB, IFN-γ/STAT1	Midstream	Mouse/comparative; inferred in human
Immunopathology	Cytokine-driven tissue injury	Downstream	Animal + clinical inference
Clinical	Fever, cytopenias, transaminitis, HLH	Most downstream	Human clinical

Cell types involved (CL): neutrophil (CL:0000775), granulocyte (CL:0000094), monocyte (CL:0000576), macrophage (CL:0000235). **Anatomical structures (UBERON):** blood (UBERON:0000178), bone marrow (UBERON:0002371), liver (UBERON:0002107), spleen (UBERON:0002106), lung (UBERON:0002048), kidney (UBERON:0002113). **Subcellular (GO CC):** nucleus (GO:0005634), mitochondrion (GO:0005739), autophagosome (GO:0005776).

Evidence Base

PMID	Title (abbrev.)	Supports finding	Evidence type
39102427	<i>HGA — systematic review</i>	F1, F4, F12 (clinical phenotype, doxycycline, prognosis)	Human clinical (systematic review)
17275372	<i>Immune evasion & immunosuppression by A. phagocytophilum</i>	F2, F10 (neutrophil subversion; ruminant TBF)	Review / in vitro
15879137	<i>Fails to induce apoptosis in human neutrophils</i>	F2 (apoptosis delay)	In vitro (human PMN microarray)
25996657	<i>Chromatin-bound Anka recruits HDAC1</i>	F3 (AnkA epigenetic silencing)	In vitro / molecular
20174550	<i>Ats-1 imported into mitochondria, blocks apoptosis</i>	F3 (Ats-1 anti-apoptosis)	In vitro
23197835	<i>Autophagosomes induced by bacterial Beclin 1-binding protein</i>	F3 (Ats-1 autophagy hijack)	In vitro
40176262	<i>Trends in anaplasmosis over the past decade</i>	F4 (prognostic risk factors)	Human clinical (465-case cohort)
18485118	<i>PSGL-1-independent infection, Syk, Anka delivery</i>	F5 (entry receptors)	In vitro
22859615	<i>Antigen variability during chronic reservoir infection</i>	F5 (msp2/p44 antigenic variation)	In vivo (reservoir host)
41016325	<i>Small mammal hosts of zoonotic A. phagocytophilum, Canada</i>	F6 (Ap-ha variant, reservoirs)	Field epidemiology
41003548	<i>Hospitalizations for TBDs in the US 2002–2021</i>	F6 (US burden)	Epidemiology (registry)
25385549	<i>Transfusion-associated infection in pregnancy</i>	F6, F9 (non-vector transmission; rifampin)	Case report
23278812	<i>IFN-γ production and Stat1 signaling</i>	F7, F12 (immunopathology)	Model organism + comparative

PMID	Title (abbrev.)	Supports finding	Evidence type
42230277	<i>CSF findings in CNS anaplasmosis</i>	F7 (cytokine-mediated, not direct CNS infection)	Human clinical
15122530	<i>TLR2 activation of NF-κB by A. phagocytophilum</i>	F7/F12 (innate sensing branch)	In vitro
20077398	<i>Ehrlichiosis/Anaplasmosis</i>	F8 (diagnostic modalities)	Review
34035378	<i>HGA in a single Korean university hospital</i>	F8 (morulae low sensitivity)	Human clinical cohort
17682993	<i>HGA during pregnancy: case series</i>	F9 (rifampin/doxycycline in pregnancy)	Case series
37803346	<i>Feline granulocytic anaplasmosis, strain typing</i>	F10 (cross-species strains)	Veterinary/molecular
36436292	<i>A. phagocytophilum in German horses</i>	F10 (equine hematologic parallels)	Veterinary retrospective
32723647	<i>Severe anaplasmosis as treatable cause of HLH</i>	F11 (HLH complication)	Case reports + review

Across the evidence base, the **clinical phenotype and treatment findings** rest on human systematic reviews and cohorts (strong for descriptive epidemiology); the **molecular subversion mechanisms** rest on robust in vitro and cell-model data; and the **immunopathology model** is strongest in mouse and comparative (horse) systems and is inferred — though well-supported — in humans by the dissociation between low bacterial burden and disease severity, and by cytokine-mediated CSF findings.

Section-by-Section Knowledge Base Content

1. Disease Information. HGA is an acute tick-borne bacterial infection of neutrophils. Identifiers: MONDO:0005118; MeSH "Anaplasmosis"/"Ehrlichiosis, Human, Granulocytic"; ICD-10 A79.82 (Anaplasmosis); ICD-11 ~1C30.2. No OMIM/Orphanet genetic entry (not a

Mendelian disease). Synonyms: human granulocytic ehrlichiosis (HGE, historical), granulocytic anaplasmosis. Information is derived from aggregated disease-level clinical and epidemiological resources, not germline genetics.

2. Etiology. Causal factor is **infectious** — the Ap-ha variant of *A. phagocytophilum*. Risk factors are environmental/behavioral: residence or activity in *Ixodes*-endemic regions (Northeast/Upper Midwest/Pacific US), outdoor exposure in tick season (peak July), advanced age, and immunosuppression (severity). **No genetic risk, protective, or gene-environment factors are established** in humans.

3. Phenotypes. Fever (88.5%; HP:0001945), thrombocytopenia (71.8%; HP:0001873), transaminitis (66.7%; HP:0002910), leukopenia (49.8%; HP:0001882); myalgia, headache, chills, arthralgia common; complications include acute renal failure (9.8%), multi-organ failure (7.5%), ARDS (6.3%). Rare: subdural hematoma/CNS involvement, pulmonary embolism, HLH. Onset acute; severity mild-to-severe/variable; course self-limited with treatment.

4. Genetic/Molecular Information. Not applicable at the human host level — no causal genes, pathogenic variants, modifier genes, or chromosomal abnormalities. The relevant molecular biology is the *pathogen's* virulence genes (ankA, ats-1, msp2/p44, T4SS/virB-virD4) and the *host* genes they target (CYBB/gp91phox silencing; mitochondrial respiratory-chain subunits NDUFB5/NDUFB3/NDUFS7/COX6C/SLC25A5 up-regulated).

5. Environmental Information. Infectious agent: *A. phagocytophilum* (NCBI Taxon 948). Vectors: *Ixodes scapularis*, *I. pacificus* (also *I. ricinus* in Europe). Reservoirs: *Peromyscus leucopus*, chipmunks, other small mammals. Transmission mainly by tick bite; rare via blood transfusion and perinatally.

6. Mechanism/Pathophysiology. See the ordered causal chain and schematic above. Molecular pathways: TLR2/NF- κ B, IFN- γ /JAK-STAT1; host NADPH-oxidase pathway (silenced); mitochondrial apoptosis and autophagy (subverted). Immune involvement is central and immunopathologic.

7. Anatomical Structures Affected. Primary: blood/hematopoietic system (neutrophils; UBERON:0000178, bone marrow UBERON:0002371). Secondary: liver (UBERON:0002107), spleen, kidney, lung, occasionally CNS. Subcellular: host cytoplasmic vacuole, nucleus (AnkA target; GO:0005634), mitochondria (Ats-1 target; GO:0005739), autophagosome (GO:0005776).

8. Temporal Development. Onset acute, ~5–14 days post-tick bite; disease is typically self-limited and resolves rapidly with doxycycline (fever ≤ 1 day). Untreated disease can progress to complications; chronic human infection is not established (unlike persistent reservoir-host infection).

9. Inheritance and Population. No inheritance (infectious). Epidemiology: rising US incidence; concentrated in Northeast/Upper Midwest; seasonal July peak; slight male predominance (53.9%); older adults over-represented among hospitalized. Not a genetic disease — penetrance/expressivity/founder effects N/A.

10. Diagnostics. Whole-blood **PCR** (primary, sensitive early), **IFA serology** (paired acute/convalescent; insensitive early), Giemsa **blood smear morulae** (specific but ~5.6% sensitive early). Supportive labs: CBC (thrombocytopenia, leukopenia), elevated transaminases, elevated CRP. Differential: ehrlichiosis, Lyme disease, babesiosis, other acute febrile illnesses.

11. Outcome/Prognosis. Good: lethality ~3.0%, sequelae ~2.1%. Complications in ~40.5%. Prognostic factors: advanced age, immunosuppression, altered mental status, comorbidities. Rapid recovery with timely doxycycline.

12. Treatment. **Doxycycline** first-line, all ages (NCIT:C731); **rifampin** in pregnancy/intolerance (NCIT:C692). Severe/HLH cases: doxycycline + corticosteroids ± anakinra (IL-1RA). No gene/cell/RNA therapies applicable.

13. Prevention. No vaccine. Primary prevention = tick-bite avoidance (permethrin-treated clothing, DEET/IR3535 repellents, tick checks, landscape/rodent-targeted interventions). Secondary prevention = early empiric treatment. Public health: vector/reservoir control, health education.

14. Other Species / Natural Disease. Natural disease in horses (equine granulocytic anaplasmosis), dogs, cats, and ruminants (tick-borne fever with abortion, reduced milk yield, immunosuppression). Shared strains across humans/dogs/horses/cats confirm zoonotic cross-species susceptibility.

15. Model Organisms. Mouse models (used to establish IFN- γ /STAT1 immunopathology); naturally infected horses, sheep/lambs (msp2/p44 persistence studies), and reservoir woodrats/*Peromyscus* (antigenic variation). In vitro: HL-60 promyelocytic cells, human PMNs, HEK293T (effector studies). These recapitulate neutrophil infection, cytopenias, and immunopathology; they do not fully capture human clinical heterogeneity.

Limitations and Knowledge Gaps

- **Immunopathology in humans is inferred, not directly demonstrated.** The IFN- γ /STAT1 model is strongest in mouse and horse systems; direct human tissue-level causal evidence is limited, resting on the burden–severity dissociation and cytokine-mediated CSF findings.
 - **Host genetic determinants of human severity are unknown.** Why some patients (beyond age/immunosuppression) develop severe disease or HLH is not defined; no human susceptibility loci have been mapped.
 - **Mechanistic link from subversion to specific cytopenias is incomplete.** The pathways producing thrombocytopenia and leukopenia (marrow suppression vs peripheral consumption/sequestration) are not fully resolved.
 - **Effector biology gaps.** Only 3 of ≥ 6 T4SS effectors are functionally characterized; the full effector repertoire and its integration with host signaling remain open.
 - **Diagnostic performance data are heterogeneous.** Sensitivity/specificity of PCR vs serology across disease stages come from varied, sometimes small cohorts.
 - **No controlled trial data** exist for HLH-directed immunomodulation in anaplasmosis; evidence is anecdotal (case reports).
 - **Publication/case-report bias** likely inflates the apparent frequency of rare severe manifestations (CNS, HLH, thromboembolism) relative to the true population.
-

Proposed Follow-up Experiments / Actions

1. **Human immunophenotyping study:** longitudinal serum cytokine profiling (IFN- γ , IL-1, IL-6, IL-18, ferritin) and pSTAT1 in circulating leukocytes in HGA patients stratified by severity, to directly test the immunopathology model and identify HLH-risk biomarkers.
2. **Host-genetics of severity:** targeted or exome sequencing of severe/HLH HGA cases (e.g., PRF1, UNC13D, STXBP2 HLH genes) to test whether occult HLH predisposition underlies severe outcomes.
3. **Effector–pathway mapping:** complete the functional characterization of the remaining T4SS effectors and their host targets using proximity labeling/interactomics in primary human neutrophils.

4. **Mechanistic dissection of cytopenias:** bone marrow and platelet-kinetic studies (or murine models) to determine whether thrombocytopenia/leukopenia are marrow-suppressive or consumptive.
 5. **Diagnostics benchmarking:** prospective multicenter head-to-head evaluation of PCR vs IFA vs smear vs emerging antigen/metagenomic assays across defined time-since-onset windows.
 6. **Immunomodulation registry:** structured case registry (or adaptive platform) for anaplasmosis-associated HLH to evaluate doxycycline ± corticosteroids ± anakinra outcomes.
 7. **Prevention translation:** field evaluation of rodent-targeted acaricide/vaccine interventions specifically on Ap-ha (human-active) variant prevalence, not just total *A. phagocytophilum*.
-

Consensus Answer

Human granulocytic anaplasmosis (HGA; MONDO:0005118) is an acute, usually self-limiting tick-borne febrile illness caused by the obligate intracellular bacterium *Anaplasma phagocytophilum* (human-active "Ap-ha" variant), transmitted mainly by *Ixodes* ticks and characterized by fever with thrombocytopenia, transaminitis, and leukopenia. The pathogen uniquely infects neutrophils — entering via PSGL-1/sialyl-Lewis^x and surviving via Type IV secretion effectors Anka (HDAC1-mediated epigenetic silencing of the NADPH-oxidase gene *CYBB*) and Ats-1 (mitochondrial anti-apoptosis and autophagy hijacking) — while tissue injury is driven by an IFN- γ /STAT1 immunopathologic host response rather than direct bacterial damage. HGA has no human genetic etiology or vaccine; diagnosis is by whole-blood PCR and IFA serology, doxycycline is rapidly curative first-line therapy (rifampin in pregnancy), and prognosis is good (lethality ~3%), with advanced age and immunosuppression predicting severe disease.
